Refugee Inflows and Native Parity Progression

Evidence from the Syrian inflow and Turkish DHS

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Why this question, why Turkey.

- Global fertility is below replacement (TFR < 2.1) in nearly all high- and middle-income countries; raising native fertility is now an explicit policy goal in many.
- Turkey's TFR **fell from 2.10 in 2010 to ≈ 1.62 in 2022** — below replacement, below the OECD average — while at the same time becoming the **largest refugee host** ($\approx$3.6M Syrians, post-2011).
- Turkey is therefore a uniquely sharp setting to study the **interaction between low native fertility and a large migration shock**.

Question. How did the post-2011 inflow affect native Turkish women's marriage and fertility outcomes **by parity and by age**?

Why parity-by-age. Two competing mechanisms make distinct predictions:

- **Marriage-market crowd-out** $\Rightarrow$ delays *first births*, depresses *marriage*, effects in early 20s.
- **Opportunity-cost / household-production** $\Rightarrow$ lowers cost of additional births where childcare/domestic labour is cheaper, effects at *higher parities, late 20s–30s*.

A pooled total-fertility regression cannot distinguish these. **The parity-age grid can.**

Data: Turkish Demographic and Health Surveys (TDHS)

Wave	Sample	Controls	Province assignment
2003	Ever-married only	Limited	Childhood province only
2008	Ever-married only	Limited	Childhood province only
2013	Ever-married *and* never-married	Full	2011 residence (mig. history)
2018	Ever-married *and* never-married	Full	2011 residence (mig. history)

Two analytical samples:

- **Primary** — 2013/18, full sample (incl. never-married), province in 2011. Most accurate pre-shock province assignment, no pre-selection on marriage.
- **Four-wave** — 2003/08/13/18, ever-married only, childhood province. Anchors trend estimation with pre-shock cohorts; restricted by data availability in 2003/08.

Today's results. Headline pooled results presented on the primary sample; four-wave estimates and decomposition ladder in the appendix.

Outcomes and Exposure

Cumulative parity outcomes (absorbing-state indicators):

$$Y_{ija}^{(p)} = \mathbb{1}\{\text{parity}_i \geq p \text{ by age } a\}, \qquad p \in \{1,2,3,4,5\}, \quad a \in \{16,20,24,28,32,36\}.$$

Plus marriage analogue $\mathbb{1}\{\text{ever married by age } a\}$.

Exposure: average refugee-to-native ratio over the woman's age window from 12 to a:

$$\bar{R}_{ja} = \frac{1}{a-11} \sum_{s=12}^{a} R_{j,\,t(s,i)}.$$

Interpretation. A 1 pp. higher *average annual* ratio over ages 12–a raises $\Pr(\text{parity} \geq p \text{ by } a)$ by $\hat{\beta}^{(p,a)}$ pp.

Cohort dependence. Two women aged a in the same province in different waves get different $\bar{R}_{ja}$ — their averaging windows cover different calendar years. This cross-cohort variation drives identification.

Shift-Share Instrument

Province-year instrument (Aksu, Erzan and Kırdar 2022):

$$Z_{n,t} = \sum_{s=1}^{13} \underbrace{\left[\frac{(1/d_{s,T})\,\pi_s}{\sum_{X\in\{T,L,J,I\}}(1/d_{s,X})} \right]}_{\text{share weight (time-invariant)}} \cdot \frac{T_t}{d_{n,s}}$$

π_s pre-war pop. share of Syrian province s; $d_{s,X}$ distance from s to host $X \in \{T, L, J, I\}$; $d_{n,s}$ distance from s to Turkish province n; T_t aggregate Syrian refugee stock across the four host countries.

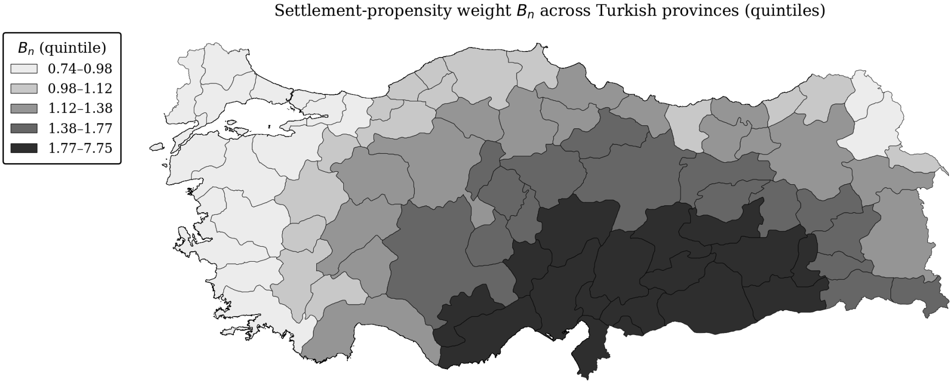
Key decomposition — only T_t varies in time:

$$Z_{n,t} = B_n \times T_t, \qquad \bar{Z}_{ja} = B_j \times \bar{T}_a.$$

$\Longrightarrow$ **Continuous-treatment difference-in-differences**: B_n assigns province intensity (geography), T_t activates the shock after 2011.

Exclusion restriction collapses to one condition: B_n uncorrelated with province-specific *pre-existing trends* in fertility, conditional on FE and trend controls.

Settlement-Propensity Weight B_n



Estimating Equation

For each (p, a) separately,

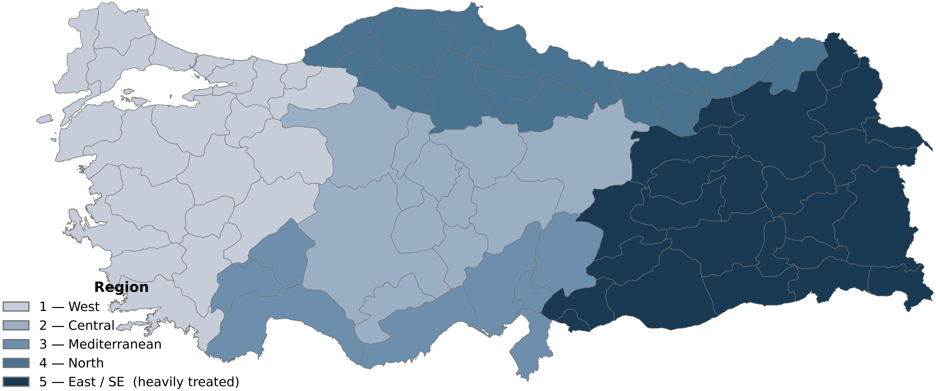
$$Y_{ija}^{(p)} = \alpha + \beta^{(p,a)}\bar{R}_{ja} + X_i'\Gamma + \delta_j + \mu_{t(a,i)} + \theta_{g(j)} \cdot t(a,i) + \lambda_{l(i)} \cdot t(a,i) + \varepsilon_{ija},$$

with $\bar{R}_{ja}$ instrumented by $\bar{Z}_{ja} = B_j \times \bar{T}_a$.

- δ_j — province FE $\quad \mu_{t(a,i)}$ — year-at-age-a FE ($\equiv$ birth-cohort FE at fixed a)
- $\theta_{g(j)} \cdot t$ — **five-region**-specific linear time trends
- $\lambda_{l(i)} \cdot t$ — **mother-tongue**-specific linear time trends (Turkish / Kurdish / Arabic)
- Trends start in **1987**; X_i: wave, childhood urban/rural, mother's education
- Standard errors clustered at province, DHS sampling weights applied

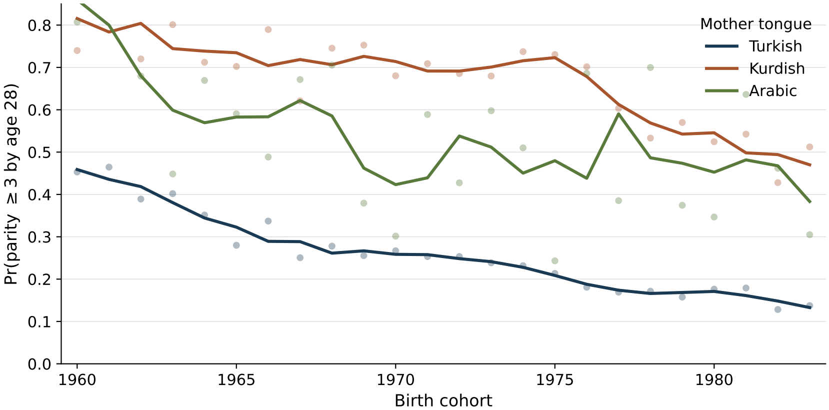
We motivate the two trend controls (region and mother tongue) on the *next two slides*: each absorbs a pre-existing differential trend that is correlated with B_n by geography or ethnicity.

Why Region-Specific Trends? — The 5-Region Scheme



Region 5 (East/SE) houses **≈95% of refugees**, has **higher baseline fertility**, and was on a **distinct demographic transition** relative to the rest of the country pre-2011 — without region-specific trends, $B_n \times T_t$ would absorb these dynamics. Five regions chosen for parsimony (NUTS-1 / NUTS-2 in appendix).

Why Mother-Tongue-Specific Trends?



Pre-shock cohorts only. **Levels and trends differ sharply** across MT groups — and Region 5 (high B_n) is **55% Kurdish**-speaking versus 1.7–12% in regions 1–4. So B_n correlates with MT composition by geography, and MT-specific trends are needed to absorb the differential transition.

Placebo: $Y \sim B_n \times t$ on Pre-Shock Cohorts (Romano–Wolf)

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	0.0003	0.0010	0.0012	0.0008	0.0011	-0.0001
	(0.0005)	(0.0007)	(0.0007)	(0.0004)	(0.0006)	(0.0007)
Parity $\geq$ 1	0.0005	0.0011	0.0007	0.0011	0.0011	0.0001
	(0.0003)	(0.0008)	(0.0010)	(0.0005)	(0.0005)	(0.0007)
Parity $\geq$ 2	0.0001	-0.0001	0.0009	0.0018	0.0015	0.0006
	(0.0002)	(0.0004)	(0.0006)	(0.0007)	(0.0010)	(0.0011)
Parity $\geq$ 3	0.0000	-0.0000	0.0004	0.0000	-0.0007	-0.0024
	(0.0000)	(0.0002)	(0.0007)	(0.0008)	(0.0008)	(0.0013)
Parity $\geq$ 4	–	–	0.0002	0.0011	0.0005	-0.0005
	(–)	(–)	(0.0004)	(0.0009)	(0.0023)	(0.0022)
Parity $\geq$ 5	–	–	0.0000	0.0002	0.0012	0.0018
	(–)	(–)	(0.0002)	(0.0005)	(0.0014)	(0.0022)
Observations	12,355	11,560	10,226	8,194	6,063	4,098

Test. On the subsample $\{i : t(a,i) < 2012\}$, regress each outcome on $B_n \times t(a,i)$ with the full set of FE and trends. After Romano–Wolf stepdown across the 32-cell family, **no cell rejects**. First-stage $F \in [43, 144]$ (FS); raw + four-wave appendix.

Main Results — Primary Sample (Romano–Wolf)

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	-0.0467	-0.5715	0.5130	-0.0130	0.2202	-0.2993
	(0.2165)	(0.6000)	(1.6892)	(1.0473)	(0.9200)	(1.2412)
Parity ≥ 1	0.0095	0.1226	-0.0431	0.9235	-1.5150	0.1839
	(0.1273)	(0.5231)	(1.9677)	(1.3419)	(1.7893)	(1.3711)
Parity ≥ 2	0.0285	0.3048	-0.1508	0.6126	1.7403	-0.8450
	(0.0409)	(0.5464)	(0.7457)	(1.9059)	(3.4194)	(1.1891)
Parity ≥ 3	0.0084	0.3816	-0.3712	1.2502	2.8929	1.0620
	(0.0055)	(0.2481)	(0.5649)	(1.5588)	(2.6999)	(1.8097)
Parity ≥ 4	–	–	0.2496	1.9047	2.2458	4.2520
	(–)	(–)	(0.3399)	(0.8094)	(1.4146)	(1.9632)
Parity ≥ 5	–	–	0.3133	1.3921	1.4136	4.1326
	(–)	(–)	(0.2324)	(0.8025)	(0.9119)	(1.8555)
Observations	14,354	13,484	12,241	10,240	8,245	6,110

Pooled point estimates positive at parities 4–5, ages 28/36 (raw $p < 0.05$ — raw), but **no cell survives Romano–Wolf** across the 32-test family ($p_{RW} \geq 0.85$ for every cell). **Signal lives in heterogeneity** (next two). Marriage / parities 1–3 null in every spec.

Four-wave (EM)

Heterogeneity I — Heavily-Treated Region (Romano–Wolf)

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	-0.2452	-1.6915	-2.5905	-0.9127	-0.4301	1.5648
	(0.3582)	(0.6464)	(1.9762)	(1.6556)	(0.7606)	(0.8548)
Parity ≥ 1	-0.0233	-0.4039	-3.0301	1.4459	0.3391	1.9888
	(0.1541)	(0.4780)	(1.5690)	(2.2608)	(1.4282)	(1.1851)
Parity ≥ 2	0.0210	1.4250	1.2829	0.4696	5.4953	0.3303
	(0.0518)	(0.4706)	(0.9335)	(2.1259)	(2.4693)	(1.1054)
Parity ≥ 3	-0.0151	0.3390	-0.0495	5.1117	9.1213**	4.4002
	(0.0193)	(0.4285)	(0.5988)	(2.0445)	(2.8160)	(1.7627)
Parity ≥ 4	–	–	0.2292	2.9874	6.7512**	5.7527
	(–)	(–)	(0.5746)	(1.2826)	(1.6983)	(2.4905)
Parity ≥ 5	–	–	-0.0136	1.8753	4.1483	6.5940
	(–)	(–)	(0.4266)	(0.9337)	(1.3901)	(3.4988)
Observations	4,002	3,522	3,022	2,404	1,851	1,311

Region 5 under Romano–Wolf: **parity ≥ 3 @ 32 $= +9.12$ pp (**)** and **parity ≥ 4 @ 32 $= +6.75$ pp (**)**.
Region 1–4 entirely null (region 1–4).

Heterogeneity II — Mother Tongue (Romano–Wolf)

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	0.5562	0.5199	0.5854	-4.0251	1.9332	0.8593
	(0.3233)	(1.7594)	(2.4777)	(2.2391)	(2.2384)	(1.1457)
Parity $\geq$ 1	0.3356	1.8205	-1.7541	-1.5852	2.4411	0.7623
	(0.1769)	(1.5930)	(2.3234)	(2.3963)	(2.7699)	(1.8112)
Parity $\geq$ 2	0.0471	1.8055	1.4326	1.4601	4.4175	1.9937
	(0.0729)	(1.1665)	(1.8375)	(2.9870)	(3.2879)	(1.4012)
Parity $\geq$ 3	-0.0192	0.6474	-0.8687	12.3048**	6.0191	2.5694
	(0.0258)	(1.2286)	(1.2296)	(2.4693)	(3.4792)	(1.5223)
Parity $\geq$ 4	–	–	1.2013	8.1188	2.0236	8.2440*
	(–)	(–)	(1.3175)	(1.7515)	(3.8106)	(1.7884)
Parity $\geq$ 5	–	–	0.3944	3.1548	4.7186	5.0376
	(–)	(–)	(1.0175)	(2.4496)	(3.3471)	(2.3657)
Observations	3,145	2,785	2,415	1,951	1,489	1,055

Kurdish/Arabic, RW: **parity $\geq$ 3 @ 28 = +12.30 (**)**, **parity $\geq$ 4 @ 36 = +8.24 (*)**. Turkish null (Turkish). Low-educ, village (low educ village) match in point estimates, lose stars. **Common thread:** **higher-baseline-fertility** groups — room to move up.

Decomposition: Where in the Parity Ladder?

Discrete-time hazard models, current-year exposure $R_{j(i,t),t}$.
For each transition k:

$$h_{ijt}^{(k)} = \alpha + \beta^{(k)} R_{j(i,t),t} + X_i'\Gamma + \phi_{\text{age}}^{(k)} + \mathbb{1}\{k \geq 2\}\psi_{\tau}^{(k)} + \delta_j + \mu_t + \theta_{g(j)} t + \lambda_{l(i)} t + \varepsilon_{ijt}^{(k)}.$$

Transition	Marriage	1st birth	2nd birth	3rd birth	4th birth	5th birth
$R_{j(i,t),t}$	0.0396 (0.0587)	-0.0502 (0.0442)	-0.0336 (0.1876)	0.0698 (0.0784)	-0.0024 (0.0661)	0.0162 (0.0646)
Person-yrs (N)	–	–	–	–	–	–

Reading. Pooled hazards are **imprecise zeros** (all $p_{RW} > 0.74$). Two non-contradictory reasons: (i) the hazard model *averages over years and women*; the parity-by-age pooled primary results in the main table are also null — the signal lives in subgroups, where subgroup hazards are work in progress; (ii) hazards are *conditional* on reaching the prior parity, so they answer "does exposure shift the 4th-birth hazard given 3 births already?" — a different question from "does exposure raise the share *ever* at parity $\geq$ 4" estimated by the Stage 3 absorbing-state outcomes.

Cross-sectional 2SLS, four-wave EM sample, current-year exposure.
Outcome: $\mathbb{1}\{\text{ideal number of children} \geq k\}$ for $k \in \{2, 3, 4, 5\}$.

Outcome	Ideal $n \geq 2$	Ideal $n \geq 3$	Ideal $n \geq 4$	Ideal $n \geq 5$
$\bar{R}_{j(i),t(i)}$	0.0458 (0.0605)	0.1739 (0.2536)	0.5142 (0.2700)	0.2555 (0.1087)
Observations	14,775	14,775	14,775	14,775

Romano–Wolf adjusted across the 4-test family. Raw point estimates: Pr(ideal ≥ 5) shifts by $+0.26$ pp per pp of average ratio (raw $p < 0.05$); ≥ 4 by $+0.51$ (raw $p < 0.10$); $\geq 2, \geq 3$ flat. Under RW $\{0.68, 0.68, 0.27, 0.21\}$, **nothing survives correction in the pooled EM sample**, but the **point-estimate pattern aligns with realised fertility**: preferences shift up at the same parities. **Subgroup ideal-n does survive RW for Kurdish/Arabic and Region 5** — see appendix.

Takeaway

The Syrian inflow raised native parity progression — in groups with higher baseline fertility.

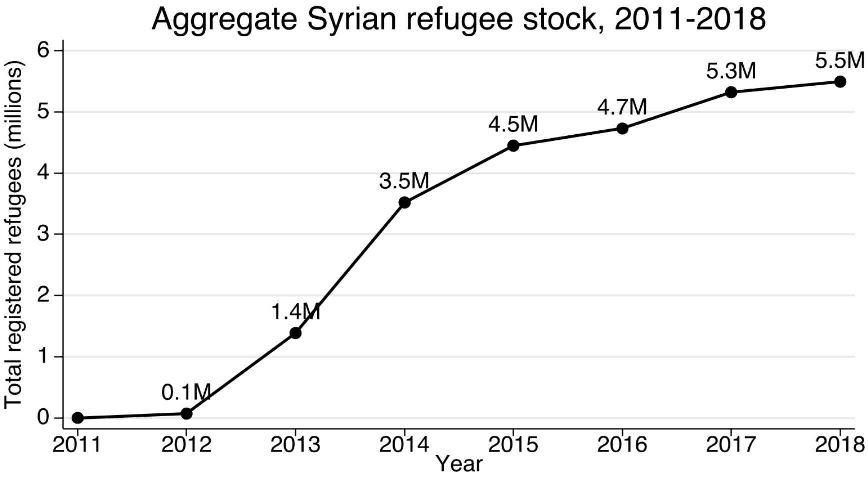
- Pooled primary positive at parities 4–5, ages 28/36 (raw stars), but **no cell survives Romano–Wolf** across the 32-test family.
- Under **Romano–Wolf** stepdown, sharp signal in heterogeneity:
  - **Region 5**: parity ≥ 3 @ $32 = +9.12\,\text{pp}$ (**); parity ≥ 4 @ $32 = +6.75\,\text{pp}$ (**).
  - **Kurdish/Arabic**: parity ≥ 3 @ $28 = +12.3\,\text{pp}$ (**); parity ≥ 4 @ $36 = +8.24\,\text{pp}$ (*).
- Marriage and first births **unaffected** everywhere — rules out marriage-market crowd-out as the dominant mechanism.
- Subjective **ideal-n**: ≥ 4, ≥ 5 shift up in the pooled (raw); RW-significant in Region 5 and Kurdish/Arabic subgroups.

Mechanism. Consistent with an **opportunity-cost / household-production** channel: refugee inflows lowered the price of household services and informal childcare, easing additional births — response concentrated where there is room to move up the parity ladder.

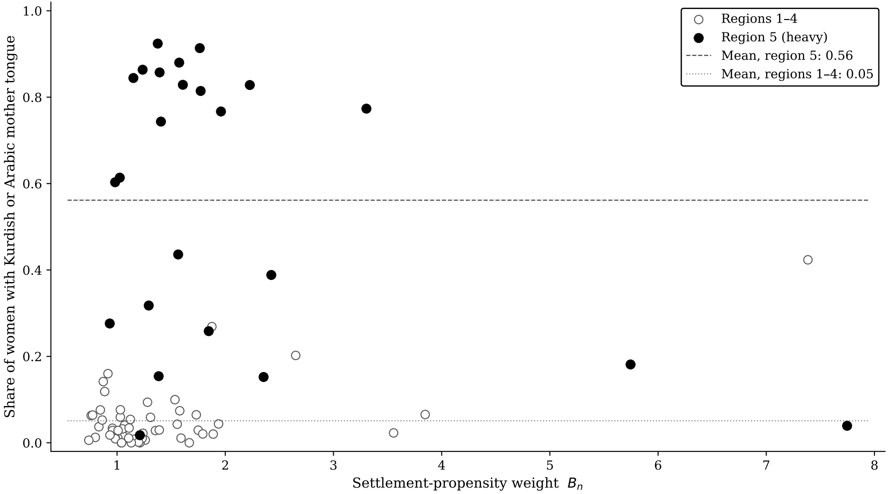
Thank you. / Questions?

Appendix

Appendix: Aggregate Refugee Stock T_t



Appendix: B_n vs Kurdish/Arabic Share at Province Level



Each dot is one province. Provinces with high B_n are also those with high Kurdish/Arabic shares — motivating the MT-specific trend control.

Appendix: Main IV — Primary Sample (Raw Analytical p)

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	-0.0467	-0.5715	0.5130	-0.0130	0.2202	-0.2993
	(0.2165)	(0.6000)	(1.6892)	(1.0473)	(0.9200)	(1.2412)
Parity ≥ 1	0.0095	0.1226	-0.0431	0.9235	-1.5150	0.1839
	(0.1273)	(0.5231)	(1.9677)	(1.3419)	(1.7893)	(1.3711)
Parity ≥ 2	0.0285	0.3048	-0.1508	0.6126	1.7403	-0.8450
	(0.0409)	(0.5464)	(0.7457)	(1.9059)	(3.4194)	(1.1891)
Parity ≥ 3	0.0084	0.3816	-0.3712	1.2502	2.8929	1.0620
	(0.0055)	(0.2481)	(0.5649)	(1.5588)	(2.6999)	(1.8097)
Parity ≥ 4	–	–	0.2496	1.9047**	2.2458	4.2520**
	(–)	(–)	(0.3399)	(0.8094)	(1.4146)	(1.9632)
Parity ≥ 5	–	–	0.3133	1.3921*	1.4136	4.1326**
	(–)	(–)	(0.2324)	(0.8025)	(0.9119)	(1.8555)
Observations	14,354	13,484	12,241	10,240	8,245	6,110

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Appendix: Main IV — Primary Sample (BH-FDR Adjusted)

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	-0.0467	-0.5715	0.5130	-0.0130	0.2202	-0.2993
	(0.2165)	(0.6000)	(1.6892)	(1.0473)	(0.9200)	(1.2412)
Parity $\geq$ 1	0.0095	0.1226	-0.0431	0.9235	-1.5150	0.1839
	(0.1273)	(0.5231)	(1.9677)	(1.3419)	(1.7893)	(1.3711)
Parity $\geq$ 2	0.0285	0.3048	-0.1508	0.6126	1.7403	-0.8450
	(0.0409)	(0.5464)	(0.7457)	(1.9059)	(3.4194)	(1.1891)
Parity $\geq$ 3	0.0084	0.3816	-0.3712	1.2502	2.8929	1.0620
	(0.0055)	(0.2481)	(0.5649)	(1.5588)	(2.6999)	(1.8097)
Parity $\geq$ 4	–	–	0.2496	1.9047	2.2458	4.2520
	(–)	(–)	(0.3399)	(0.8094)	(1.4146)	(1.9632)
Parity $\geq$ 5	–	–	0.3133	1.3921	1.4136	4.1326
	(–)	(–)	(0.2324)	(0.8025)	(0.9119)	(1.8555)
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← Back

Appendix: Main IV — Four-Wave (EM), Romano–Wolf

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Parity ≥ 1	0.3132	-1.3481	0.9856	-0.6077	-1.3975	1.3735
	(1.9142)	(1.2286)	(2.2382)	(1.1088)	(1.4367)	(0.6797)
Parity ≥ 2	0.3222	1.2727	0.6867	-1.0745	2.2526	0.0998
	(0.1720)	(2.2397)	(1.4517)	(2.0183)	(2.6152)	(1.9274)
Parity ≥ 3	0.1429	1.4820	-0.7720	-0.9003	0.7142	-0.5607
	(0.1168)	(0.9743)	(0.9236)	(1.9910)	(2.3653)	(2.5075)
Parity ≥ 4	–	–	-0.3685	0.4007	0.0293	2.7419
	(–)	(–)	(0.5178)	(0.7539)	(1.5816)	(2.0043)
Parity ≥ 5	–	–	0.2797	0.7892	-0.3457	1.5672
	(–)	(–)	(0.3048)	(0.5506)	(1.2736)	(1.6244)
Observations	19,281	21,525	21,782	19,776	16,442	12,282

← Back

Appendix: Main IV — Four-Wave (EM), Raw

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Parity ≥ 1	0.3132	-1.3481	0.9856	-0.6077	-1.3975	1.3735**
	(1.9142)	(1.2286)	(2.2382)	(1.1088)	(1.4367)	(0.6797)
Parity ≥ 2	0.3222*	1.2727	0.6867	-1.0745	2.2526	0.0998
	(0.1720)	(2.2397)	(1.4517)	(2.0183)	(2.6152)	(1.9274)
Parity ≥ 3	0.1429	1.4820	-0.7720	-0.9003	0.7142	-0.5607
	(0.1168)	(0.9743)	(0.9236)	(1.9910)	(2.3653)	(2.5075)
Parity ≥ 4	–	–	-0.3685	0.4007	0.0293	2.7419
	(–)	(–)	(0.5178)	(0.7539)	(1.5816)	(2.0043)
Parity ≥ 5	–	–	0.2797	0.7892	-0.3457	1.5672
	(–)	(–)	(0.3048)	(0.5506)	(1.2736)	(1.6244)
Observations	19,281	21,525	21,782	19,776	16,442	12,282

Appendix: Placebo — Primary Sample (Raw)

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	0.0003	0.0010	0.0012*	0.0008**	0.0011*	-0.0001
	(0.0005)	(0.0007)	(0.0007)	(0.0004)	(0.0006)	(0.0007)
Parity $\geq$ 1	0.0005*	0.0011	0.0007	0.0011**	0.0011*	0.0001
	(0.0003)	(0.0008)	(0.0010)	(0.0005)	(0.0005)	(0.0007)
Parity $\geq$ 2	0.0001	-0.0001	0.0009	0.0018**	0.0015	0.0006
	(0.0002)	(0.0004)	(0.0006)	(0.0007)	(0.0010)	(0.0012)
Parity $\geq$ 3	0.0000	-0.0000	0.0004	0.0000	-0.0007	-0.0024*
	(0.0000)	(0.0003)	(0.0007)	(0.0009)	(0.0008)	(0.0014)
Parity $\geq$ 4	–	–	0.0002	0.0011	0.0005	-0.0005
	(–)	(–)	(0.0004)	(0.0009)	(0.0023)	(0.0022)
Parity $\geq$ 5	–	–	0.0000	0.0002	0.0012	0.0018
	(–)	(–)	(0.0002)	(0.0005)	(0.0014)	(0.0023)
Observations	12,355	11,560	10,226	8,194	6,063	4,098

← Back

Appendix: Placebo — Four-Wave (EM), BH

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Parity $\geq$ 1	0.0003	0.0004	0.0003	0.0008	0.0006	0.0002
	(0.0002)	(0.0005)	(0.0005)	(0.0004)	(0.0002)	(0.0002)
Parity $\geq$ 2	0.0001	0.0002	0.0004	0.0003	0.0014	0.0005
	(0.0001)	(0.0003)	(0.0003)	(0.0005)	(0.0005)	(0.0003)
Parity $\geq$ 3	0.0000	0.0002	-0.0002	-0.0009	-0.0007	-0.0008
	(0.0000)	(0.0001)	(0.0004)	(0.0004)	(0.0003)	(0.0006)
Parity $\geq$ 4	–	–	-0.0004	-0.0011	-0.0013	-0.0016
	(–)	(–)	(0.0002)	(0.0004)	(0.0008)	(0.0009)
Parity $\geq$ 5	–	–	0.0002	-0.0009	-0.0013	-0.0024*
	(–)	(–)	(0.0001)	(0.0003)	(0.0008)	(0.0006)
Observations	18,943	20,519	20,204	17,909	14,367	10,350

Appendix: First Stage — Primary and Four-Wave

Primary sample:

	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
First-stage F	43.93	131.31	102.39	43.72	52.90	143.70
Observations	14,354	13,484	12,241	10,240	8,245	6,110

Four-wave (EM) sample:

	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
First-stage F	29.40	32.91	51.24	43.49	53.01	83.21
Observations	19,281	21,525	21,782	19,776	16,442	12,282

The four-wave specification differs from the primary in three ways. We isolate each by running the IV on four nested samples:

	Waves	Sample	Province	Isolates
(i)	2013/18	Full	2011 res.	**primary**
(ii)	2013/18	Ever-married	2011 res.	EM-restriction (i)→(ii)
(iii)	2013/18	Ever-married	Childhood	Childhood-prov. ME (ii)→(iii)
(iv)	2003/08/13/18	Ever-married	Childhood	Added-waves (iii)→(iv)

Tables on the next four slides — fertility outcomes only.

Appendix: Decomposition Rung (i) — Primary, raw

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Parity ≥ 1	0.0095	0.1226	-0.0431	0.9235	-1.5150	0.1839
	(0.1273)	(0.5231)	(1.9677)	(1.3419)	(1.7893)	(1.3711)
Parity ≥ 2	0.0285	0.3048	-0.1508	0.6126	1.7403	-0.8450
	(0.0409)	(0.5464)	(0.7457)	(1.9059)	(3.4194)	(1.1891)
Parity ≥ 3	0.0084	0.3816	-0.3712	1.2502	2.8929	1.0620
	(0.0055)	(0.2481)	(0.5649)	(1.5588)	(2.6999)	(1.8097)
Parity ≥ 4	–	–	0.2496	1.9047**	2.2458	4.2520**
	(–)	(–)	(0.3399)	(0.8094)	(1.4146)	(1.9632)
Parity ≥ 5	–	–	0.3133	1.3921*	1.4136	4.1326**
	(–)	(–)	(0.2324)	(0.8025)	(0.9119)	(1.8555)
Observations	14,354	13,484	12,241	10,240	8,245	6,110

Appendix: Decomposition Rung (ii)

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Parity $\geq$ 1	0.4099	-0.4872	0.9206	-0.8402	-2.4637	0.2145
	(1.7248)	(1.0891)	(2.6042)	(1.2393)	(1.5712)	(0.6479)
Parity $\geq$ 2	0.1292	1.4537	1.2401	-0.5771	1.6371	-0.8876
	(0.1952)	(2.1300)	(1.3208)	(1.9762)	(3.3087)	(1.4882)
Parity $\geq$ 3	0.0359	1.2429	-0.1566	0.7115	2.9164	1.3860
	(0.0922)	(0.9420)	(0.8372)	(1.6160)	(2.7133)	(2.2519)
Parity $\geq$ 4	–	–	-0.1431	1.7677**	2.2596	4.6098*
	(–)	(–)	(0.4861)	(0.8790)	(1.3754)	(2.3812)
Parity $\geq$ 5	–	–	0.2610	1.3126	1.4374	4.3275**
	(–)	(–)	(0.3487)	(0.8506)	(0.9243)	(1.9093)
Observations	10,721	11,572	11,283	9,746	7,931	5,907

Appendix: Decomposition Rung (iii)

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Parity ≥ 1	-0.5138	-0.4345	1.4650	0.3480	-2.7761	-0.2932
	(2.0057)	(1.0637)	(2.7451)	(0.7849)	(1.8694)	(0.7883)
Parity ≥ 2	-0.0428	1.2401	2.3784*	-0.5379	0.1194	-1.3289
	(0.1973)	(2.2961)	(1.3101)	(1.6543)	(2.4396)	(1.4551)
Parity ≥ 3	0.0144	1.2056	-0.0581	-0.3178	-0.2215	-1.7220
	(0.0948)	(0.8786)	(0.7228)	(1.5141)	(2.0330)	(2.4169)
Parity ≥ 4	–	–	-0.2682	1.7497*	2.0977	0.8757
	(–)	(–)	(0.3732)	(0.9399)	(1.5434)	(2.6843)
Parity ≥ 5	–	–	0.0946	0.7475	1.1589	2.3016*
	(–)	(–)	(0.3453)	(0.7486)	(1.2384)	(1.2486)
Observations	10,721	11,572	11,283	9,746	7,931	5,907

Appendix: Decomposition Rung (iv) — 4-Wave

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Parity ≥ 1	-0.6658	-1.0503	1.4667	0.2834	-2.4309	0.9380
	(2.2741)	(1.1459)	(2.3698)	(0.6135)	(1.9752)	(0.6439)
Parity ≥ 2	0.1760	1.0901	1.9913	-1.0349	0.7577	0.2081
	(0.1972)	(2.3668)	(1.3790)	(1.4748)	(2.2178)	(1.2591)
Parity ≥ 3	0.1306	1.3511	-0.5372	-1.0839	-0.0772	-0.9134
	(0.1237)	(0.9421)	(0.8202)	(1.6311)	(2.0124)	(1.9287)
Parity ≥ 4	–	–	-0.3500	1.2114	1.1983	0.6504
	(–)	(–)	(0.3757)	(1.0065)	(1.5172)	(1.8778)
Parity ≥ 5	–	–	0.2083	1.2281	1.0989	0.6201
	(–)	(–)	(0.2610)	(0.9975)	(1.0670)	(1.2463)
Observations	19,281	21,525	21,782	19,776	16,442	12,282

Appendix: Region 1–4 ("Other"), Primary (BH)

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	-0.2609	-1.3578	1.7261	0.0472	-2.0559	-1.9054
	(0.1868)	(1.0728)	(1.7837)	(1.2570)	(1.6061)	(0.7543)
Parity $\geq$ 1	0.0452	-0.3939	1.8234	1.3042	-6.0697	-0.9893
	(0.2257)	(1.0197)	(1.4001)	(2.0493)	(2.5172)	(0.9780)
Parity $\geq$ 2	-0.0363	-0.5946	-1.2016	2.6652	-4.8945	-1.1427
	(0.0429)	(0.6160)	(0.9670)	(2.1099)	(3.1711)	(1.6412)
Parity $\geq$ 3	-0.0006	0.1853	-0.6541	-0.1352	0.1920	0.5938
	(0.0014)	(0.3480)	(0.7896)	(3.9169)	(3.8745)	(2.6857)
Parity $\geq$ 4	–	–	0.2275	1.4477	1.4941	6.1002
	(–)	(–)	(0.4780)	(1.3774)	(2.3666)	(2.5449)
Parity $\geq$ 5	–	–	0.5797	0.8921	1.4853	5.4374
	(–)	(–)	(0.2703)	(1.3467)	(1.6360)	(2.7984)
Observations	10,352	9,962	9,219	7,836	6,394	4,799

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Appendix: Turkish, Primary (BH)

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	-0.4330	-1.2864	-0.1425	0.2402	-1.5190	-2.3851
	(0.1970)	(0.7963)	(1.6814)	(1.6210)	(1.2953)	(2.9491)
Parity $\geq$ 1	-0.1718	-0.8128	-0.5693	1.2331	-3.9264	-2.1423
	(0.1202)	(0.7729)	(2.3789)	(1.9282)	(2.7780)	(3.3247)
Parity $\geq$ 2	-0.0034	0.0618	-1.2228	0.1915	-0.7790	-3.6633
	(0.0239)	(0.7466)	(0.9056)	(2.7639)	(5.4613)	(2.3983)
Parity $\geq$ 3	0.0041	0.1653	-0.3431	-1.0986	2.7682	-0.8493
	(0.0043)	(0.3170)	(0.5697)	(1.7848)	(4.0056)	(3.3966)
Parity $\geq$ 4	–	–	-0.0748	-0.3461	2.5819	1.8207
	(–)	(–)	(0.3250)	(1.0642)	(1.0607)	(2.9586)
Parity $\geq$ 5	–	–	0.2606	-0.0430	0.0264	5.1158
	(–)	(–)	(0.1146)	(0.3103)	(1.5401)	(3.8226)
Observations	11,013	10,512	9,655	8,137	6,634	4,960

← Back

Appendix: Low Education, Primary (BH)

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	-0.1848	-0.8722	2.0480	-0.0590	0.1893	-0.2025
	(0.3031)	(1.2854)	(1.8239)	(1.5294)	(1.0346)	(1.1901)
Parity ≥ 1	-0.0234	0.0484	0.2907	1.3034	0.0446	0.5583
	(0.1870)	(1.2489)	(2.4427)	(1.9572)	(1.2737)	(1.3313)
Parity ≥ 2	0.0029	0.2609	-1.1164	1.6909	1.6018	1.9856
	(0.0571)	(1.6357)	(1.6824)	(1.6453)	(2.7430)	(1.6354)
Parity ≥ 3	0.0038	0.7375	-0.6375	2.8227	3.2607	4.6894
	(0.0069)	(0.7647)	(1.2099)	(2.2384)	(2.4464)	(1.6967)
Parity ≥ 4	–	–	-0.0667	3.1747	2.9678	6.6579
	(–)	(–)	(0.6786)	(1.2937)	(1.0527)	(2.3695)
Parity ≥ 5	–	–	0.3507	1.7869	1.8934	5.6194
	(–)	(–)	(0.4137)	(1.2837)	(1.1604)	(1.8604)
Observations	9,361	9,132	8,729	7,633	6,348	4,897

← Back

Appendix: High Education, Primary (BH)

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	0.0176	0.2143	-0.8859	-1.4122	-1.6326	1.9265
	(0.0534)	(0.5891)	(3.0142)	(3.2686)	(2.9202)	(4.8016)
Parity $\geq$ 1	0.0414	-0.3907	-0.5787	0.4812	-9.4252	0.6246
	(0.0289)	(0.6715)	(2.9107)	(3.7083)	(4.9289)	(5.4977)
Parity $\geq$ 2	0.0006	0.3035	0.2737	0.1005	-4.5550	-15.0970
	(0.0235)	(0.2388)	(1.1549)	(4.6549)	(7.5530)	(8.5615)
Parity $\geq$ 3	nan	0.0801	0.0441	-3.2287	7.3840	-18.7862
	(nan)	(0.0614)	(0.8062)	(1.2656)	(6.9828)	(7.4317)
Parity $\geq$ 4	–	–	-0.5710	-0.3215	3.9271	-8.8020
	(–)	(–)	(0.4974)	(0.3705)	(3.1890)	(4.8629)
Parity $\geq$ 5	–	–	-0.4857	0.1304	-0.3653	-2.3138
	(–)	(–)	(0.4814)	(0.1855)	(0.5670)	(2.1605)
Observations	4,993	4,352	3,512	2,607	1,897	1,213

Appendix: Childhood Village, Primary (BH)

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	0.1211	-1.4415	0.8995	1.0317	1.0009	3.4578***
	(0.2736)	(1.1784)	(1.5356)	(1.4479)	(0.9845)	(0.6143)
Parity $\geq$ 1	0.1742	-0.1255	0.4177	2.6297	0.5035	3.3643
	(0.1738)	(1.2916)	(2.2922)	(2.4481)	(1.5035)	(0.8700)
Parity $\geq$ 2	0.1081	0.7126	-1.9560	3.8106	-1.0797	4.7298
	(0.0906)	(0.9498)	(1.8797)	(2.8029)	(4.0945)	(1.4622)
Parity $\geq$ 3	0.0114	1.1294	-1.8534	4.0855	1.8258	4.4503
	(0.0137)	(0.6116)	(2.3677)	(4.0760)	(3.2759)	(2.7838)
Parity $\geq$ 4	–	–	-0.8554	2.3753	10.8030	4.8970
	(–)	(–)	(1.2278)	(2.9829)	(5.2495)	(2.8621)
Parity $\geq$ 5	–	–	-0.0620	-0.5112	3.3327	6.2334
	(–)	(–)	(0.5388)	(1.3538)	(2.6586)	(2.0937)
Observations	5,951	5,876	5,551	4,810	4,014	3,144

← Back

Appendix: Childhood Province/District Center, Primary (BH)

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Ever Married	-0.2411	-0.3869	0.4872	-1.6019	-0.3382	-5.7737
	(0.2742)	(1.0264)	(2.2987)	(1.3923)	(1.5140)	(2.9858)
Parity $\geq$ 1	-0.1708	0.0695	0.2356	-1.1591	-3.1994	-4.2549
	(0.1566)	(0.7220)	(2.4566)	(1.3131)	(2.1961)	(3.2637)
Parity $\geq$ 2	-0.0267	0.1538	0.4950	-2.2219	2.3199	-6.5088
	(0.0362)	(0.5424)	(1.2147)	(2.4383)	(3.2030)	(2.0111)
Parity $\geq$ 3	-0.0016	-0.0079	0.4753	-0.4963	4.8564	-4.0918
	(0.0068)	(0.1930)	(1.1148)	(1.6198)	(2.6472)	(3.4030)
Parity $\geq$ 4	–	–	0.6621	1.4791	-0.4328	3.4158
	(–)	(–)	(0.4222)	(1.1368)	(1.6984)	(5.7876)
Parity $\geq$ 5	–	–	0.4354	1.9743	1.7589	3.3016
	(–)	(–)	(0.2074)	(1.0362)	(1.1231)	(3.5413)
Observations	8,403	7,608	6,690	5,430	4,231	2,966

Appendix: Heterogeneity — Region Heavy, EM (BH)

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Parity ≥ 1	3.4637	-0.1417	-0.7001	1.1050	1.2919	0.7813
	(1.6930)	(1.0351)	(0.9967)	(1.8278)	(1.1700)	(0.9819)
Parity ≥ 2	0.7230	4.9204	3.9068	-0.0815	6.4816	0.7833
	(0.3124)	(2.2036)	(2.0116)	(1.8604)	(2.1972)	(1.0608)
Parity ≥ 3	0.2263	1.1774	-0.2640	4.8317	9.5649*	4.2373
	(0.1818)	(1.3025)	(1.3782)	(2.0548)	(2.4606)	(1.6056)
Parity ≥ 4	–	–	0.5677	3.1868	7.3546	8.3427
	(–)	(–)	(1.0679)	(1.4297)	(1.8899)	(2.3085)
Parity ≥ 5	–	–	0.3315	2.4357	4.3646	7.3215
	(–)	(–)	(0.5873)	(1.1019)	(1.6014)	(3.8179)
Observations	5,666	6,103	5,823	4,995	3,954	2,796

Outcome	Age 16	Age 20	Age 24	Age 28	Age 32	Age 36
Parity ≥ 1	1.4726	-1.0009	-1.1131	-2.2038	1.7139	0.3218
	(1.2192)	(2.0293)	(2.6882)	(2.1287)	(1.8921)	(1.4880)
Parity ≥ 2	0.2515	2.8340	2.1400	0.8100	2.9979	2.0974
	(0.4117)	(1.7714)	(3.6033)	(2.4592)	(2.4310)	(1.5816)
Parity ≥ 3	-0.0823	2.0225	-1.3910	12.3488**	3.2532	1.3143
	(0.3290)	(1.9931)	(1.8745)	(2.4731)	(3.0358)	(2.2161)
Parity ≥ 4	–	–	-0.1841	8.4375	2.0872	7.1089
	(–)	(–)	(1.7918)	(2.4696)	(3.8543)	(1.7496)
Parity ≥ 5	–	–	0.3056	4.1805	6.1599	6.0199
	(–)	(–)	(0.9846)	(2.6576)	(2.8704)	(2.0907)
Observations	4,492	4,765	4,517	3,885	3,041	2,157

Appendix: Ideal Fertility — Region 5 and Kurdish/Arabic (RW)

Region 5 (heavy) — Romano–Wolf:

Outcome	Ideal $n \geq 2$	Ideal $n \geq 3$	Ideal $n \geq 4$	Ideal $n \geq 5$
$\bar{R}_{j(i),t(i)}$	0.0688 (0.1169)	0.2123 (0.2372)	0.7273* (0.3361)	0.3676** (0.1428)
Observations	4,003	4,003	4,003	4,003

Kurdish / Arabic — Romano–Wolf:

Outcome	Ideal $n \geq 2$	Ideal $n \geq 3$	Ideal $n \geq 4$	Ideal $n \geq 5$
$\bar{R}_{j(i),t(i)}$	0.1287 (0.1578)	0.5124 (0.4172)	0.9109** (0.3501)	0.4754 (0.1844)
Observations	3,177	3,177	3,177	3,177

Appendix: Ideal Fertility — Education and Childhood Residence (RW)

Low education — Romano–Wolf:

Outcome	Ideal $n \geq 2$	Ideal $n \geq 3$	Ideal $n \geq 4$	Ideal $n \geq 5$
$\bar{R}_{j(i),t(i)}$	0.0012 (0.0879)	0.2032 (0.2468)	0.6757** (0.2471)	0.3513** (0.1005)
Observations	11,723	11,723	11,723	11,723

Childhood village — Romano–Wolf:

Outcome	Ideal $n \geq 2$	Ideal $n \geq 3$	Ideal $n \geq 4$	Ideal $n \geq 5$
$\bar{R}_{j(i),t(i)}$	-0.1961 (0.1081)	-0.0944 (0.2524)	0.5148 (0.4037)	0.2867 (0.2053)
Observations	7,275	7,275	7,275	7,275